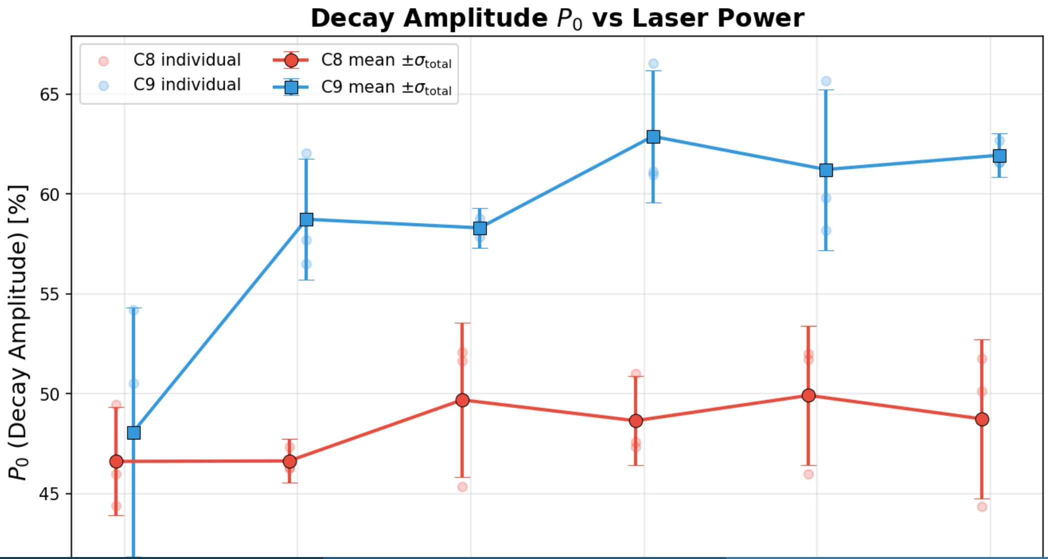
Power Dependence of MEOP Fit Parameters

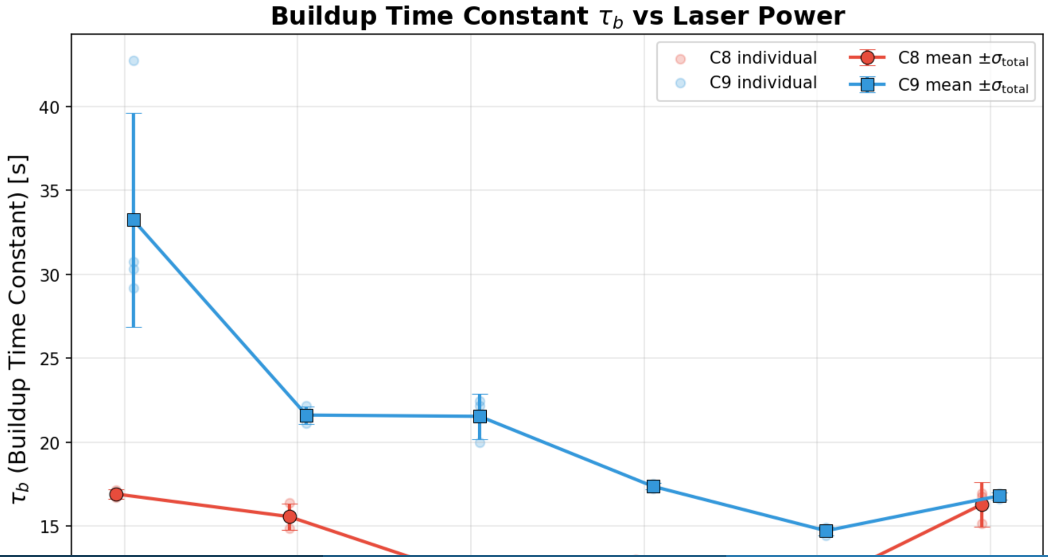
Cells C8 & C9 — 3 W to 8 W

MEOP Analysis

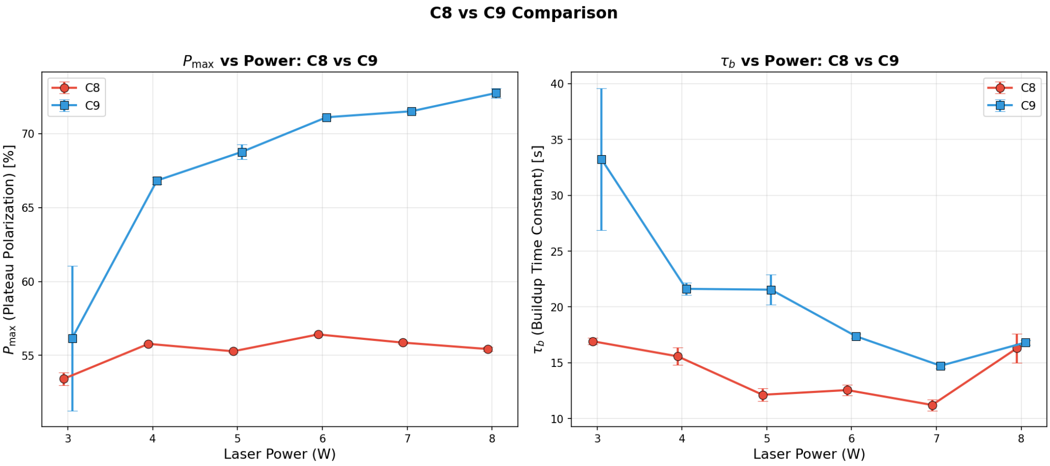
February 27, 2026


Decay Amplitude P_0 vs Laser Power
C8 individual
C9 individual
C8 mean $\pm\sigma_{total}$
C9 mean $\pm\sigma_{total}$
65
60
55
50
45
P_0 (Decay Amplitude) [%]

Buildup Time Constant τ_b vs Laser Power



C8 vs C9 Comparison



$\sigma_{\text{total}} = \sqrt{\bar{\sigma}_{\text{fit}}^2 + \sigma_{\text{rms}}^2}$, $\bar{\sigma}_{\text{fit}} = \sqrt{\sum \sigma_{\text{fit},i}^2 / N}$

- C8
- C9

Error: $\sigma_{\mathrm{fit}} \oplus \sigma_{\mathrm{rms}}$ added in quadrature

Summary of Observations

Decay Amplitude P_0

- C8: Relatively flat (∼47–50 %) across all powers
- C9: Increases from ∼48 % (3 W) to ∼63 % (6 W), then plateaus
- C9 consistently higher than C8 above 4 W

Buildup Time τ_b

- Both cells: τ_b decreases with increasing power (faster buildup)
- C9 starts much higher (∼33 s at 3 W) vs C8 (∼17 s)
- Convergence at high power (∼16 s at 8 W)

All error bars: $\sigma_{\text{total}} = \sqrt{\sigma_{\text{fit}}^2 + \sigma_{\text{rms}}^2}$ (fit uncertainty $\oplus$ run-to-run spread, independent errors)